

Table 5. Event-Level Response Modeling Summary

Model family	Best model	Accuracy	Balanced acc.	Macro F1	Top-3 acc.	JS div.	Interpretation
Majority baseline	Majority	0.203	0.100	0.034	0.523	0.6055	Lower-bound class-prior baseline
Best actor-only	Hist-GBDT actor only	0.166	0.104	0.096	0.441	0.0731	Actor word alone is weak
Best actor-history exact/top-3	XGBoost actor history	0.470	0.403	0.429	0.677	0.0163	Best autonomous candidate response prediction
Best actor-history distributional	Random Forest actor history	0.440	0.408	0.398	0.629	0.0025	Best actor-history distributional
Best observed-immediate exact	XGBoost observed immediate	0.771	0.739	0.779	0.861	0.0068	Best offline exact response
Best observed-immediate distributional	Random Forest observed immediate	0.759	0.751	0.731	0.823	0.0022	Best observed-immediate distributional